

Class - 15

SIGNIFICANCE LEVEL & ERRORS

HYPOTHESIS TESTING

Significance Level

The **significance level (α)** is the **cut-off point** we use to decide whether to reject H_0 .

It tells us **how much chance we are willing to take to make a wrong decision.**

Let's take an example to understand **significance level**

If $\alpha = 0.05$

This means $\rightarrow$ "I am okay with being wrong **5 times out of 100** decisions I make."

So, when we reject the null hypothesis (H_0), there's still a **small chance (5%)** that we might be making a mistake.

Type 1 and Type 2 Errors

We saw when we were doing the hypothesis testing we are having null hypothesis and alternate hypothesis.

and at the end we have to check whether we accept the null hypothesis or reject the null hypothesis.

now we have reality -

where either the null hypothesis is true or it is false.

And then we have decision -

where we will decide the null hypothesis is true or false.

Type 1 and Type 2 Errors

So based on this information we will have 4 outcomes :

- 1) We reject the null hypothesis in reality it was false.
- 2) We reject the null hypothesis in reality it was true. —————→ **Type 1 error**
- 3) We accept the null hypothesis in reality it was True.
- 4) We accept the null hypothesis in reality it was false. —————→ **Type 2 error**

The Confusion matrix for Type 1
and Type 2 Error is -

	P	N	
T	TP	TN	Type 1 error
F	FP	FN	

Type 2 error

Type 1 and Type 2 Errors

1. Type 1 Error :

Type 1 error happens when we **reject H_0 even though it's actually true.**

It's like sounding a false alarm.

Let's take an example for this -

Assume we have a court case

H_0 : The person is innocent.

If we wrongly say "Guilty!" when the person was actually innocent.

(Now that's Type 1 error)

Type 1 and Type 2 Errors

2. Type 2 Error :

This happens when we **fail to reject H_0 even though it's false.**

It's like missing something that's actually there.

Let's take an example for this -

Assume we have a court case

H_0 : The person is innocent.

If we say "Not Guilty" but in reality the person was guilty.

(Now that's Type 2 error)

ONE TAILED AND TWO TAILED TEST

1. One-Tailed Test :

A **one-tailed test** is a type of hypothesis test where the **rejection region** (the area where we reject H_0) is placed **all in one tail** of the probability distribution, either on the **left side** or on the **right side**.

In simple words,

The One-Tailed Test is used when we care about **only one difference**, either greater or smaller.

It's like One-tailed test is saying → "I only care about one side , either more or less."

→ All of α (significance level) goes to **one tail**.

ONE TAILED AND TWO TAILED TEST

2. Two-Tailed Test :

A **two-tailed test** is a type of hypothesis test where the **rejection region** is split into **both tails** of the probability distribution one on the **left side** and one on the **right side**.

In simple words,

The Two-Tailed Test is used when we care about **any difference**, whether greater **or** smaller.

So, the “rejection region” is **split into two parts** — one on the **left tail** and one on the **right tail**.

P - Value

The p-value is the probability of getting results as extreme as (or more extreme than) the sample result **if the null hypothesis (H_0) is true**.

In simple words :

It tells us **how surprising our data is** under the assumption that H_0 is true.

Smaller p-value = stronger evidence **against H_0** .

There is a Decision Rule in P-value -

- If **p-value $< \alpha$ (significance level)** $\rightarrow$ Reject H_0 .
- If **p-value $\geq \alpha$** $\rightarrow$ Fail to reject H_0 .

P - Value

Let's see an example for P-value,

Suppose a company claims 50% of customers like their product.

We take a sample and calculate a P-value

For now let's say, $P\text{-value} = 0.03 / 3\%$

Interpretation:

- If H_0 were true (50% like it), there's only a **3% chance** of getting results this extreme.
- and Since $0.03 < 0.05$ ($\alpha = 5\%$), the data is **too rare** to believe H_0 .

Decision:

Reject $H_0 \rightarrow$ More customers like the product than claimed.

Test Statistics

A **test statistic** is a special number we calculate from our sample data to help us decide whether to reject the **Null hypothesis (H_0)**.

It measures **how far** our sample result is from what we would expect if **H_0** were true.

Then we compare this number to critical values (or use the p-value) to make a decision.

- Examples of test statistics:
 - **Z-test**
 - **t-test**
 - **Chi-square**
 - **F-test**